

9 Elliptic integrals

9.1 Introduction

The integral of any rational function is expressible in terms of rational functions and logarithms. The same functions, augmented by square roots, suffice to integrate

$$\int r(x, y) dx, \quad (1)$$

where r is any rational function of x and y (i.e. the ratio of two polynomials in x and y) and y is the square root of a linear or quadratic polynomial in x . (We interpret logarithms in the sense of Example 6.9-4 to include inverse circular and inverse hyperbolic functions.) However, if y^2 is a cubic or quartic polynomial, (1) is not in general expressible in terms of elementary functions. Integrals of this type were encountered very early in the history of calculus, for example, in finding the length of an arc of an ellipse, and hence they came to be called elliptic integrals. They arise also in connection with the arc length of a hyperbola or a Bernoulli lemniscate, the surface area of an ellipsoid, the electric capacity and demagnetizing factors of ellipsoidal bodies, the motion of a pendulum and other kinds of anharmonic oscillators, the rotation of a top, the conformal mapping of a half-plane onto a rectangle (Example 8.2-2), and many other problems of applied and pure mathematics.

More generally let

$$y^2 = p_n(x), \quad (2)$$

where p_n is a polynomial of exact degree n . The nature of the integral (1) depends more critically on p_n , and especially on n , than on the detailed nature of the rational function r . If $n > 4$, then (1) is called a hyperelliptic integral (provided p_n has no multiple zero and r is not even in y). The surface area of an ellipsoid in $\mathbb{R}^d$ is a hyperelliptic integral if $d \geq 4$, and so is its capacity if $d \geq 5$, but hyperelliptic integrals have much less practical importance than elliptic integrals.

The integrals considered so far are special cases of two quite different classes. The class of Abelian integrals consists of integrals of the form (1) in which y is an algebraic function of x . This means that y satisfies

$$p(x, y) = 0, \quad (3)$$

where p is a polynomial in x and y . In the real domain, (3) may be regarded geometrically as a curve in the plane, and the theory of Abelian integrals is intimately connected with the theory of algebraic curves. [See Erdélyi (1953, Section 13.2), Siegel (1969), and Hille (1959, Chapter 12).] Except in the elementary and elliptic cases, Abelian integrals do not have major importance in applied mathematics. Even the simple integrals representing the beta function (4.2-2) and Gauss's hypergeometric function (Exercise 4.4-2) are not Abelian unless the parameters are rational numbers.

The second class consists of integrals of the form

$$\int (A + Bx)^\alpha \cdots (M + Nx)^\mu dx, \quad (4)$$

in which the integrand is a product of powers of linear factors ($A, B, \alpha, \dots, M, N, \mu$ being possibly complex constants). Since every polynomial is a product of linear factors, the integral of any rational function of x has the form (4) with exponents $\alpha, \dots, \mu$ that are integers. Moreover, if y is given by (2), any integral of type (1) is the sum of two integrals of type (4). Since even powers of y are polynomials in x , $r(x, y)$ can be written as

$$r(x, y) = \frac{a(x) + yb(x)}{c(x) + yd(x)}, \quad (5)$$

where a, b, c, d are polynomials. Multiplying and dividing by $c - yd$, we obtain

$$r(x, y) = \rho(x) + y\sigma(x), \quad (6)$$

where ρ and σ are rational functions. The integral of each term has the form (4) since y^2 is a product of n linear factors. In the integral of the second term exactly n of the exponents $\alpha, \dots, \mu$ are $\frac{1}{2}$ while the rest are integers. Thus the class of integrals of the form (4) contains the logarithmic, elliptic, and hyperelliptic classes. Since it contains also the integrals

representing the beta function and Gauss's hypergeometric function, (4) is more pertinent to applied mathematics than (1) and (3).

By the formulas of Section 8.1 the integral (4) is expressible as an R function (multiplied by a beta function), and the exponents $\alpha, \dots, \mu$ coincide (except for order) with all but one of the parameters $a - 1, a' - 1, -b_1, -b_2, \dots$, the remaining parameter being determined by the condition $a + a' = \sum b_i$. If n of the exponents $\alpha, \dots, \mu$ are half-odd integers while the rest are integers, and if h of the parameters $a, a', b_1, b_2, \dots$ are half-odd-integers while the rest are integers, then $h = n$ if n is even and $h = n + 1$ if n is odd. In particular the integral of a rational function ($n = 0$) is an R function with $h = 0$, while $h = 2$ if $n = 1$ or $n = 2$. Theorems 8.5-1 and 8.5-4 justify the statements made about such integrals at the beginning of this section. An elliptic integral ($n = 3$ or $n = 4$) is an R function with $h = 4$ if we disregard the R function with $h = 0$ coming from the first term on the right side of (6).

A change in perspective is suggested by the condition $a, a' \in \mathbb{C}_>$ which is necessary for convergence of the integrals (8.1-2) to (8.1-4). Use of those formulas to evaluate any convergent elliptic integral of the form (4) always leads to an R function with positive values of a and a' . On the other hand, to take an example, all members of the set of associated functions

$$R_{1/2-m}(\tfrac{1}{2} + m, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z), \quad m \in \mathbb{Z}, \quad (7)$$

are basically similar in nature regardless of the sign of m . This suggests that "elliptic integrals" might better be regarded as a class of functions than a class of integrals, specifically the class of R functions such that four of the parameters $a, a', b_1, \dots, b_k$ (with k unrestricted) are half-odd integers and the rest are integers.

The change in viewpoint is a matter of emphasis. It has an important influence on what seems simple and natural, for instance in deciding what questions to ask and in choosing standard functions, but does not alter fundamentally the notion of an elliptic integral. We illustrate this by (7) with $m = 0$. The function

$$R_{1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z) \quad (8)$$

is symmetric in x, y, z and has simple representations as a double integral (5.9-1) and an infinite series (5.9-4). As a double integral it expresses the surface area of an ellipsoid (Example 9.4-2). Because $a = -\frac{1}{2}$, it cannot be represented by the single integral (6.8-2) and will not arise naturally from (4) or (1). With some effort it can be maneuvered into the form (1) by using (5.9-6) to represent it as a linear combination of functions with $a = \frac{1}{2}$,

each admitting the representation (6.8-6). The result is

$$R_{1/2}\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; x, y, z\right) = \frac{1}{4} \int_0^\infty [(t+x)(t+y)(t+z)]^{-1/2} \cdot \left(\frac{x}{t+x} + \frac{y}{t+y} + \frac{z}{t+z}\right) t \, dt. \quad (9)$$

The right side is an elliptic integral in the conventional sense, but it is not a natural or convenient representation of the left side. From the point of view that elliptic integrals are a class of R functions, the traditional definition by (1) is unnatural and restrictive because it stresses too much a particular representation by a single integral. Although the subject of elliptic integrals began as a problem in integration, to regard it this way no longer seems advantageous.

The first major properties of elliptic integrals, the addition theorems, were discovered by Euler in the period 1752–1761, and the theory was brought to completion in many respects by Legendre in his treatise of 1825–1826. During the years 1827–1830 Abel and Jacobi discovered the elegant properties of elliptic functions, which overshadowed elliptic integrals for the rest of the nineteenth century. Elliptic functions are the inverses of certain elliptic integrals (see Section 8.2), just as the exponential function is the inverse of the logarithm and the sine function is the inverse of the arcsine. For example consider the integrals

$$y = \int_1^x t^{-1} \, dt = \log x, \quad z = \int_0^x (1-t^2)^{-1/2} \, dt = \arcsin x, \\ u = \int_0^x (1-t^2)^{-1/2} (1-k^2 t^2)^{-1/2} \, dt, \quad (10)$$

where k is a possibly complex constant. The integrals y, z, u are many-valued functions of the complex variable x and can be written by using (8.1-2) as R functions with $h = 0, 2, 4$ respectively. In the first two cases the inverse functions, obtained by regarding the upper limit of integration as a function of the integral, are $x = e^y$ and $x = \sin z$. They are single-valued and periodic. In the third case the inverse function is single-valued and doubly periodic; that is, the complex u plane can be divided by two sets of parallel lines into congruent parallelograms such that the elliptic function $x(u)$ has exactly the same behavior in every parallelogram (see Fig. 8.2-1, in which the crosses are vertices of one parallelogram). In the degenerate case when the parallelograms stretch to infinite strips, one of the two fundamental periods becomes infinite, and the elliptic function reduces to a simply periodic circular or hyperbolic function.

From the third equation of (10) we find the differential equation

$$\begin{aligned} du/dx &= (1-x^2)^{-1/2}(1-k^2x^2)^{-1/2}, \\ (x')^2 &= (1-x^2)(1-k^2x^2) = 1 - (1+k^2)x^2 + k^2x^4, \end{aligned} \quad (11)$$

where a dash signifies differentiation with respect to u . A second differentiation gives

$$x'' + (1+k^2)x = 2k^2x^3. \quad (12)$$

In comparison with elliptic integrals, elliptic functions are simple because they are single-valued but at the same time more complicated because they satisfy nonlinear differential equations like (12). Elliptic integrals satisfy linear homogeneous partial differential equations [(5.4-2) and the second equation of (5.9-2)] because they are R functions, that is, hypergeometric functions of several variables. Although Appell invented hypergeometric functions of more than one variable in 1880, their connection with elliptic integrals went unnoticed for eighty years [see Carlson (1961)].

9.2 Symmetric standard functions

As noted in the last section, elliptic integrals are the class of R functions such that four of the parameters $a, a', b_1, \dots, b_k$ are half-odd integers and the rest are integers. By Theorem 8.5-3 elliptic integrals are expressible in terms of rational functions, square roots, and functions of four variables of the form $R_{-\alpha}(\beta, w)$, where $\alpha, \beta_1, \beta_2, \beta_3$ are half-odd integers while α' and β_4 are integers (α' being defined by $\alpha + \alpha' = \sum_{i=1}^4 \beta_i$). Between any five such functions there exists (by Theorem 8.4-3) a linear homogeneous relation with polynomial coefficients. If we choose four standard functions that are linearly independent with respect to polynomial coefficients, any such function can be expressed as a linear combination of the four standard functions with coefficients that are rational functions.

There are infinitely many ways to choose four standard functions, but some choices are clearly better than others. If β_4 is 0, it can be eliminated together with w_4 by (6.3-3). We make this choice for three of the functions but not for the fourth, because four functions of three variables would be linearly dependent by Theorem 8.4-3. A negative integer is likewise ruled out because it can be raised to zero and eliminated by (5.9-7). Hence we choose $\beta_4 = 1$ in the fourth standard function.

In all four functions we choose $\beta_1 = \beta_2 = \beta_3$ to achieve symmetry in w_1, w_2, w_3 . Any half-odd integer will work, but $\beta_1 = \beta_2 = \beta_3 = \frac{1}{2}$ seems the most natural choice. It is in fact the best choice because it occurs in

the function which maps a half-plane onto the interior of a rectangle (Example 8.2-2) and has a doubly periodic inverse function.

Since all β parameters have now been chosen, α is determined by choosing the integer α' . By (6.8-15) a function with $\alpha' \leq 0$ is the product of $(w_1 w_2 w_3)^{-1/2}$ and a rational function. We choose $\alpha' = 0$ for one function, but linear independence then requires α' to be positive, say 1 or 2, for the others. Relabeling w_1, w_2, w_3, w_4 as x, y, z, ρ , we have now arrived at four standard functions (not yet proved to be linearly independent),

$$\begin{aligned} (xyz)^{-1/2} &= R_{-3/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z), \\ R_F(x, y, z) &= R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z), \\ R_G(x, y, z) &= R_{1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z), \\ R_H(x, y, z, \rho) &= R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, 1; x, y, z, \rho). \end{aligned} \quad (1)$$

The abbreviation R_F was explained in Example 8.2-2, and G and H follow alphabetically. We have already met R_G in (9.1-8). In the first three functions the α' parameter is 0, 1, 2. There are good arguments for choosing $\alpha' = 1$ in R_H , but we have chosen $\alpha' = 2$ to give the function a simpler numerical behavior in the real domain as ρ tends to 0 or ∞ .

An elliptic integral is called complete if both a and a' are half-odd integers. In proving linear independence it will be helpful to consider the complete integrals

$$\begin{aligned} R_K(x, y) &= R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}; x, y), \\ R_E(x, y) &= R_{1/2}(\tfrac{1}{2}, \tfrac{1}{2}; x, y), \\ R_L(x, y, \rho) &= R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, 1; x, y, \rho). \end{aligned} \quad (2)$$

By Theorem 8.3-2 these are the cases of R_F , R_G , and R_H in which one variable vanishes:

$$\begin{aligned} R_F(x, y, 0) &= (\pi/2)R_K(x, y), \\ R_G(x, y, 0) &= (\pi/4)R_E(x, y), \\ R_H(x, y, 0, \rho) &= (3\pi/8)R_L(x, y, \rho). \end{aligned} \quad (3)$$

We have already met R_K in Example 6.10-2, in Sections 8.2–8.3 and in Exercise 6.10-4, where it represented the period of a simple pendulum. The function R_E represents the circumference of an ellipse [see (9.4-5)]. Applying Theorem 8.3-2 a second time, we find

$$R_E(x, 0) = (2/\pi)x^{1/2}, \quad R_L(x, 0, \rho) = (4/\pi)R_C(x, \rho), \quad (4)$$

where R_C is the logarithmic function discussed in Example 6.9-14. By (3) and (4),

$$R_H(x, 0, 0, \rho) = \frac{3}{2}R_C(x, \rho). \quad (5)$$

Since even this special case of R_H has a logarithmic dependence on ρ , R_H is linearly independent of the first three standard functions with respect to coefficients that are rational in x, y, z, ρ .

To prove the linear independence of the first three functions, we consider their behavior for positive x, y, z as $z \rightarrow \infty$. By (3) and (4),

$$R_G(x, 0, 0) = \frac{1}{2}x^{1/2}, \quad (6)$$

and therefore

$$R_G(x, y, z) = z^{1/2}R_G\left(\frac{x}{z}, \frac{y}{z}, 1\right) \sim z^{1/2}R_G(0, 0, 1) = \frac{1}{2}z^{1/2}, \quad z \rightarrow \infty. \quad (7)$$

For R_F we use (6.8-2),

$$\begin{aligned} z^{1/2}R_F(x, y, z) &= R_F\left(\frac{x}{z}, \frac{y}{z}, 1\right) \\ &= \frac{1}{2} \int_0^1 u^{-1/2} \left(1 - u + u \frac{x}{z}\right)^{-1/2} \left(1 - u + u \frac{y}{z}\right)^{-1/2} du. \end{aligned} \quad (8)$$

Setting $u^{-1/2} = u^{-1/2}(1 - u^{1/2}) + 1$, we find

$$\begin{aligned} z^{1/2}R_F(x, y, z) &= I_1 + I_2, \\ I_1 &= \frac{1}{2} \int_0^1 u^{-1/2}(1 - u^{1/2}) \left(1 - u + u \frac{x}{z}\right)^{-1/2} \left(1 - u + u \frac{y}{z}\right)^{-1/2} du, \\ I_2 &= \frac{1}{2} \int_0^1 \left(1 - u + u \frac{x}{z}\right)^{-1/2} \left(1 - u + u \frac{y}{z}\right)^{-1/2} du \\ &= \frac{1}{2} R_{-1}\left(\frac{1}{2}, \frac{1}{2}, 1; \frac{x}{z}, \frac{y}{z}, 1\right). \end{aligned} \quad (9)$$

As $z \rightarrow \infty$, by the Lebesgue theorem of dominated convergence (Appendix B.1),

$$I_1 \rightarrow \frac{1}{2} \int_0^1 u^{-1/2}(1 - u^{1/2})(1 - u)^{-1} du = \int_0^1 (1 + v)^{-1} dv = \log 2,$$

where we have substituted $u = v^2$. By Exercise 8.5-1,

$$I_2 = \frac{1}{2}zR_{-1}\left(\frac{1}{2}, \frac{1}{2}, 1; x, y, z\right) \sim \log[2z^{1/2}/(x^{1/2} + y^{1/2})], \quad z \rightarrow \infty.$$

Therefore, for positive x, y, z ,

$$R_F(x, y, z) \sim z^{-1/2} \log[4z^{1/2}/(x^{1/2} + y^{1/2})], \quad z \rightarrow \infty. \quad (10)$$

By (7) and (10), the asymptotic forms of the first three functions in (1) are linearly independent with respect to polynomial coefficients. Together with the independence of R_H already established, this completes the proof of the following theorem.

THEOREM 9.2-1 The four standard functions $(xyz)^{-1/2}$, $R_F(x, y, z)$, $R_G(x, y, z)$, and $R_H(x, y, z, \rho)$ are linearly independent with respect to coefficients that are rational in x, y, z, ρ .

The symmetric standard functions R_F , R_G , and R_H (respectively called elliptic integrals of the first, second, and third kinds) are modern replacements for the three standard functions which have been used since the end of the eighteenth century. Legendre was the first to show that only three nonelementary functions are needed, but he chose them without regard to permutation symmetry. Expressed as R functions (see Example 9.3-1), his choices were

$$\begin{aligned} F(\varphi, k) &= (\sin \varphi) R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; \cos^2 \varphi, 1 - k^2 \sin^2 \varphi, 1), \\ E(\varphi, k) &= (\sin \varphi) R_{-1/2}(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{3}{2}; \cos^2 \varphi, 1 - k^2 \sin^2 \varphi, 1), \end{aligned} \quad (11)$$

$$\Pi(\varphi, k, \alpha^2)$$

$$= (\sin \varphi) R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, 1; \cos^2 \varphi, 1 - k^2 \sin^2 \varphi, 1, 1 - \alpha^2 \sin^2 \varphi).$$

Comparison with (1) shows that F is essentially R_F , but the other two functions have less than maximal symmetry because of unequal b parameters. One independent variable has been eliminated in each case by using homogeneity, with a further sacrifice of symmetry that affects even F .

Because of their lack of symmetry Legendre's standard functions have linear transformations. Their origin is made clear by rewriting the relation (11) between R_F and F as

$$\begin{aligned} R_F(x, y, z) &= (z - x)^{-1/2} F \left[\arccos \left(\frac{x}{z} \right)^{1/2}, \left(\frac{z - y}{z - x} \right)^{1/2} \right] \\ &= (z - y)^{-1/2} F \left[\arccos \left(\frac{y}{z} \right)^{1/2}, \left(\frac{z - x}{z - y} \right)^{1/2} \right]. \end{aligned} \quad (12)$$

Since R_F is symmetric, the third member is obtained by interchanging x and y . The equality of the second and third members is a linear transformation of F , called the reciprocal-modulus transformation because it changes the modulus $k = [(z - y)/(z - x)]^{1/2}$ into $1/k$ (see Exercise 9.2-2). Similarly

the five nontrivial permutations of x, y, z induce five linear transformations of F which change k into $1/k, k', 1/k', ik/k', k'/ik$, where $k' = (1 - k^2)^{1/2}$. The linear transformations have no fundamental content and are merely the spawn of a notation which hides the permutation symmetry.

The linear transformations of the other two functions also are empty and are more complicated because of the intrinsic asymmetry coming from unequal b parameters. We shall verify in Example 9.3-1 that

$$R_G(x, y, z) = \frac{1}{2} (z - x)^{1/2} E \left[\arccos \left(\frac{x}{z} \right)^{1/2}, \left(\frac{z - y}{z - x} \right)^{1/2} \right] \\ + \frac{1}{2} x (z - x)^{-1/2} F \left[\arccos \left(\frac{x}{z} \right)^{1/2}, \left(\frac{z - y}{z - x} \right)^{1/2} \right] + \frac{1}{2} \left(\frac{xy}{z} \right)^{1/2}. \quad (13)$$

Permutations of x, y, z induce linear transformations of E which contain also F and an elementary function. It is hard to imagine that anyone would choose E and F as standard functions if he knew that the right sides of (13) and (12) are symmetric. Legendre (1786, p. 617) chose E (perhaps the letter as well as the function) because it represents the arc length of an ellipse (see Example 9.4-1). Euler and Legendre (1786, p. 616; 1825, Vol. I, pp. vii, 3) initially expected that elementary analysis using logarithms and circular arcs (i.e. inverse circular functions) would develop into a higher analysis using elliptic arcs. However, Legendre had to augment E by two more standard functions to make the foundation of a higher analysis, which consisted in the reduction of all elliptic integrals to these three functions. It was E that was defined first and set the pattern for the others, but the fundamental importance of F undermined the original emphasis on elliptic arcs. In retrospect the right generalization of a circle is not an ellipse but a torus, the domain of definition of a doubly periodic function in the same sense in which a circle is the domain of definition of a periodic function of a real variable.

Setting $\varphi = \pi/2$ in (11) yields Legendre's complete integrals,

$$K(k) = F(\pi/2, k) = \frac{\pi}{2} R_{-1/2} \left(\frac{1}{2}, \frac{1}{2}; 1 - k^2, 1 \right), \\ E(k) = E(\pi/2, k) = \frac{\pi}{2} R_{-1/2} \left(-\frac{1}{2}, \frac{3}{2}; 1 - k^2, 1 \right) \\ = \frac{\pi}{2} R_{1/2} \left(\frac{1}{2}, \frac{1}{2}; 1 - k^2, 1 \right), \\ \prod_1(k, \alpha^2) = \prod (\pi/2, k, \alpha^2) = \frac{\pi}{2} R_{-1/2} \left(\frac{1}{2}, -\frac{1}{2}, 1; 1 - k^2, 1, 1 - \alpha^2 \right). \quad (14)$$

We have used Theorem 8.3-2 and (5.9-19). Comparison with (2) shows that

$$R_K(x, y) = \frac{2}{\pi} y^{-1/2} K \left[\left(1 - \frac{x}{y} \right)^{1/2} \right], \quad R_E(x, y) = \frac{2}{\pi} y^{1/2} E \left[\left(1 - \frac{x}{y} \right)^{1/2} \right]. \quad (15)$$

Since R_K and R_E are symmetric, interchange of x and y induces a linear transformation of K and E which changes k into ik/k' .

9.3 Reduction to standard functions

Legendre showed that any elliptic integral given in the form (9.1-1) can be expressed in terms of his standard functions. He began essentially by proceeding to (9.1-6) and decomposing the rational functions ρ and σ by the method of partial fractions. The corresponding problem for us is to express any given R function with exactly four half-odd-integral parameters (the rest being integers) in terms of the standard functions (9.2-1). Since the given function might come from a double integral and have a negative a parameter, we should not start from (9.1-1). We know from Section 9.2 that the reduction is possible because of relations between associated R functions. In this section we shall describe a systematic procedure and list some typical reduction formulas to show how the theory works in practice.

The proof of Theorem 8.5-3 contains a procedure for reducing the given function to R functions of four variables in which the homogeneity parameter and b_1, b_2, b_3 are half-odd integers while b_4 is a nonnegative integer. (See Section 9.8 for direct reduction of some cases in which both a and a' are integers.) We now continue from this point and discuss first the case $b_4 = 0$. The function $R_{-a}(b_1, b_2, b_3; x, y, z)$ is to be expressed as a linear combination of the first three standard functions in (9.2-1). If a' (defined by $a + a' = b_1 + b_2 + b_3$) is a nonpositive integer, the function reduces to an algebraic function by (6.8-15). If a' is a positive integer, the function is called an elliptic integral of the second kind except in the case of R_F , the standard integral of the first kind.

The first step of a systematic procedure for reducing an integral of the second kind is to change each b parameter to $\frac{1}{2}$. A b parameter less than $\frac{1}{2}$ can be raised by successive applications of (5.9-7), and one greater than $\frac{1}{2}$ can be lowered by successive applications of Exercise 5.9-8. Any positive integer a' can then be reduced to the values 2, 1, 0, which correspond to standard functions, by successive applications of the four-term recurrence relation in Exercise 5.9-9. Since this procedure is tedious, it is useful to tabulate the cases which occur most frequently. By permutation symmetry we may assume $b_1 \leq b_2 \leq b_3$. Table 9.3-1 is limited to $a' = 1$, from which

Table 9.3-1

Elliptic integrals of the second kind^a

$2b$	C	C_G	C_F	C_A
$(-1, 1, 3)$	$z - y$	2	$-y$	$-xy$
$(-1, 3, 3)$	$(y - z)^2/3$	-4	$y + z$	$x(y + z)$
$(1, 1, 3)$	$(z - x)(z - y)/3$	-2	z	xy
$(1, 3, 3)$	$(z - y)d/5$	$2(3x - t)$	$3yz - s$	$x(t^2 - 2s - xt)$
$(3, 3, 3)$	$d^2/7$	$4(3s - t^2)$	$st - 9p$	$st^2 + 3pt - 4s^2$

^a In each case $a' = 1$ (whence $a = b_1 + b_2 + b_3 - 1$), $-1/2 \leq b_1 \leq b_2 \leq b_3 \leq 3/2$, and $a > 0$. We list the coefficients in

$$CR_{-a}(b_1, b_2, b_3; x, y, z) = C_G R_G(x, y, z) + C_F R_F(x, y, z) + C_A (xyz)^{-1/2},$$

using the abbreviations $t = x + y + z$, $s = xy + yz + zx$, $p = xyz$, $d = (x - y)(y - z)(z - x)$.

larger values of a' can be obtained by successive applications of (5.9-8). Cases with both b_1 and a negative are omitted, being of less practical interest because neither representation (5.9-1) nor (6.8-2) exists. Some cases with $b_3 = \frac{3}{2}$ or $a' > 1$ are listed by Nellis (1966). Instead of using the systematic procedure just described, it is easier to obtain the entries in Table 9.3-1 by using Exercise 5.9-8 to get the case in which $2b = (1, 1, 3)$ and deducing the rest from this by successive applications of the first equation in Exercise 5.9-6.

EXAMPLE 9.3-1 Express Legendre's integrals

$$F(\varphi, k) = \int_0^\varphi (1 - k^2 \sin^2 \theta)^{-1/2} d\theta, \quad E(\varphi, k) = \int_0^\varphi (1 - k^2 \sin^2 \theta)^{1/2} d\theta \quad (1)$$

in terms of symmetric standard functions. By Exercise 8.1-2 and permutation symmetry, we find

$$\begin{aligned} F(\varphi, k) &= (\sin \varphi) R_F(\cos^2 \varphi, 1 - k^2 \sin^2 \varphi, 1), \\ E(\varphi, k) &= (\sin \varphi) R_{-1/2}(-\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{3}{2}; 1 - k^2 \sin^2 \varphi, \cos^2 \varphi, 1), \end{aligned} \quad (2)$$

in agreement with (9.2-11). The first line of Table 9.3-1 yields

$$\begin{aligned} (\sin \varphi) E(\varphi, k) &= 2R_G(1 - k^2 \sin^2 \varphi, \cos^2 \varphi, 1) \\ &\quad - (\cos^2 \varphi) R_F(1 - k^2 \sin^2 \varphi, \cos^2 \varphi, 1) \\ &\quad - (\cos \varphi) (1 - k^2 \sin^2 \varphi)^{1/2}. \end{aligned} \quad (3)$$

Setting $\cos^2 \varphi = x/z$ and $1 - k^2 \sin^2 \varphi = y/z$ and using homogeneity and

symmetry of the standard functions, we prove also (9.2-13). (See Exercise 9.3-1 for $\prod$.)

The function $R_{-a}(b_1, b_2; x, y)$ is called a complete elliptic integral of the second kind (except for R_K , the complete integral of the first kind) if a, a', b_1, b_2 are half-odd integers with $a + a' = b_1 + b_2 > 0$. It can be expressed in terms of the standard complete integrals R_K and R_E by setting $z = 0$ in a reduction formula for an incomplete integral and using (8.3-4) and (9.2-2). Alternatively, b parameters can be raised or lowered to $\frac{1}{2}$ by successive applications of (5.9-7) or (5.9-24), and the homogeneity parameter can be raised or lowered by (5.9-25). Because of permutation symmetry we may assume $b_1 \leq b_2$, and because of (5.9-21) we may assume $a \leq a'$. Cases with $b_2 \leq \frac{3}{2}$ and $a' \leq \frac{3}{2}$ are listed in Table 9.3-2.

Table 9.3-2

Complete elliptic integrals of the second kind^a

$2b$	$2(a, a')$	C	C_E	C_K
$(-1, 3)$	$(1, 1)$	y	1	0
$(-1, 3)$	$(-1, 3)$	1	2	$-x$
$(1, 3)$	$(1, 3)$	$x - y$	-2	$2x$
$(3, 3)$	$(3, 3)$	$(x - y)^2$	-16	$8(x + y)$

^a In each case $b_1 \leq b_2 \leq 3/2$ and $a \leq a' \leq 3/2$, where $b_1 + b_2 = a + a' > 0$. The listed coefficients are those in

$$CR_{-a}(b_1, b_2; x, y) = C_E R_E(x, y) + C_K R_K(x, y).$$

The function $R_{-a}(b_1, b_2, b_3, b_4; x, y, z, \rho)$ is an elliptic integral of the third kind if b_4 and a' are positive integers while a, b_1, b_2, b_3 are half-odd integers. It suffices to consider $a' = 1$ since larger values of a' can be reduced by successive applications of (5.9-8). For example R_H , with $a' = 2$, can be expressed in terms of functions with $a' = 1$ by choosing $i = 4$ in (5.9-8) to obtain

$$2R_H(x, y, z, \rho) = 3R_F(x, y, z) - \rho R_{-3/2}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1; x, y, z, \rho). \quad (4)$$

Solving this equation for the third member provides the last line of Table 9.3-3. The other lines can then be deduced by raising or lowering b_1, b_2, b_3 in successive steps of unity, each step being accomplished by setting $j = 4$ in the first equation of Exercise 5.9-6 and using Table 9.3-1. In this way Table 9.3-3 can be extended to include any integral of the third kind with $a' = b_4 = 1$.

Table 9.3-3

Elliptic integrals of the third kind^a

$2(b_1, b_2, b_3)$	C	C_H	C_G	C_F	C_A
$(-1, -1, 3)$	$\rho(z - \rho)$	$-2(x - \rho)(y - \rho)/3$	2ρ	$xy - x\rho - y\rho$	$-xyp$
$(-1, 1, 1)$	3ρ	$2(\rho - x)$	0	$3x$	0
$(-1, 1, 3)$	$\rho(y - z)(z - \rho)/3$	$-2(x - \rho)(y - z)/3$	2ρ	$xy - xz - y\rho$	$-xyp$
$(1, 1, 1)$	ρ	-2	0	3	0

^a In each case $b_4 = a' = 1$, $-1/2 \leq b_1 \leq b_2 \leq b_3 \leq 3/2$, and $a = 1/2$ or $3/2$. Note that $a = b_1 + b_2 + b_3$ since $a' = b_4$. The listed coefficients are those in

$$CR_{-a}(b_1, b_2, b_3, 1; x, y, z, \rho) = C_H R_H(x, y, z, \rho) + C_G R_G(x, y, z) + C_F R_F(x, y, z) + C_A (xyz)^{-1/2}.$$

A table with $a' = 1$ and $b_4 = 2$ can be constructed in the same way, using Table 9.3-3 in place of Table 9.3-1. A single line from which to start is obtained by setting $t = -\frac{3}{2}$ and $b = (-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1)$ in (5.9-5). The left side is algebraic by (6.6-5), and all terms on the right except $R_{-3/2}(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 2; x, y, z, \rho)$ are reducible by Table 9.3-3. The table with $b_4 = 2$ can then be used in constructing a table with $b_4 = 3$, and so forth.

The function $R_{-a}(b_1, b_2, b_3; x, y, \rho)$ is a complete elliptic integral of the third kind if b_3 is a positive integer while a, a', b_1, b_2 are half-odd integers with $a + a' = b_1 + b_2 + b_3 > 0$. It can be expressed in terms of the standard complete integrals R_K, R_E, R_L by setting $z = 0$ in a reduction formula for an incomplete integral (Table 9.3-3) and using (8.3-4) and (9.2-2). Alternatively, the next to the last line in Table 9.3-4 follows from (5.9-8) with

Table 9.3-4

Complete elliptic integrals of the third kind^a

$2(b_1, b_2)$	$2(a, a')$	C	C_L	C_E	C_K
$(-1, 1)$	$(1, 1)$	2ρ	$\rho - x$	0	$2x$
$(-1, 3)$	$(1, 3)$	$y - \rho$	$x - \rho$	2	$-2x$
$(-1, 3)$	$(3, 1)$	$y\rho(y - \rho)$	$y(\rho - x)$	-2ρ	$2xy$
$(1, 1)$	$(3, 1)$	ρ	-1	0	2
$(1, 3)$	$(3, 3)$	$(x - y)(y - \rho)/4$	$x - y$	2	$-2x$

^a In each case $b_3 = 1$, $-1/2 \leq b_1 \leq b_2 \leq 3/2$, and $a, a' = 1/2$ or $3/2$. It is essential that $a + a' = b_1 + b_2 + 1 > 0$. The listed coefficients are those in

$$CR_{-a}(b_1, b_2, 1; x, y, \rho) = C_L R_L(x, y, \rho) + C_E R_E(x, y) + C_K R_K(x, y).$$

$i = 3$, $t = -\frac{1}{2}$, and $b = (\frac{1}{2}, \frac{1}{2}, 1)$. All functions with $b_3 = 1$ are obtained by successive applications of (5.9-8) with $i = 3$ or Exercise 5.9-6 with $j = 3$. These equations allow any two of a, a', b_1, b_2 to be held fixed while the other two change by unity, the third term in each equation being obtained from Table 9.3-2. A table with $b_3 = 2$ can be constructed in the same way, using Table 9.3-4 instead of Table 9.3-2. A single line from which to start is obtained by setting $t = -\frac{3}{2}$ and $b = (\frac{1}{2}, \frac{1}{2}, 1)$ in (5.9-5). The table with $b_3 = 2$ can then be used in constructing a table with $b_3 = 3$, and so forth.

9.4 Applications

This section contains some examples illustrating the occurrence of elliptic integrals in geometry, potential theory, and mechanics. Elliptic integrals are often found in unexpected places, but they are common in problems connected with ellipses and ellipsoids as well as in mechanics problems with nonlinear equations of motion. To see how they typically arise in mechanics, suppose that conservation of energy is expressed by

$$\frac{m}{2} \left(\frac{dx}{dt} \right)^2 + V(x) = E, \quad (1)$$

where the mass m and total energy E are constants and the potential energy V is a cubic or quartic polynomial in x . (If V were quadratic, the equation of motion obtained by differentiating with respect to t would be linear.) By integrating

$$dt/dx = (m/2)^{1/2} [E - V(x)]^{-1/2}, \quad (2)$$

we express the time t as an elliptic integral depending on x . Inversely x is an elliptic function of t . If the motion is periodic, the period is a complete elliptic integral.

We begin with some properties of ellipses and ellipsoids. Properties which are unchanged by permuting the principal axes, for example the surface area of an ellipsoid, are represented by symmetric functions like the standard functions of Section 9.2.

EXAMPLE 9.4-1 (*Arc length of an ellipse*) The problem of rectifying an elliptic arc stimulated the study of elliptic integrals in the seventeenth and eighteenth centuries. Euler and Legendre envisioned a higher analysis using elliptic arcs instead of circular arcs (see Section 9.2), and Legendre's standard function $E(\varphi, k)$, defined in (9.3-1), is a relic of that scheme. Although Landen's transformation originated in the reduction of a hyperbolic arc to two elliptic arcs, the conic sections were otherwise unrewarding.

The really fruitful problem was the rectification of Bernoulli's lemniscate (Exercise 8.3-7), which led Fagnano to a duplication theorem, Euler to the addition theorem, and Gauss to doubly periodic functions.

The ellipse $x^2/\alpha^2 + y^2/\beta^2 = 1$ is described in parametric form by $x = \alpha \cos \varphi$, $y = \beta \sin \varphi$. The arc in the first quadrant from $(\alpha, 0)$ to (x, y) has length

$$s = \int_0^\varphi (\alpha^2 \sin^2 \theta + \beta^2 \cos^2 \theta)^{1/2} d\theta = \beta \int_0^\varphi (1 - k^2 \sin^2 \theta)^{1/2} d\theta = \beta E(\varphi, k),$$

$$k = (1 - \alpha^2/\beta^2)^{1/2}. \quad (3)$$

By Exercise 8.1-2 or (9.3-2),

$$\begin{aligned} s &= \beta^2 \sin \varphi R_{-1/2}(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{3}{2}; \beta^2 \cos^2 \varphi, \alpha^2 \sin^2 \varphi + \beta^2 \cos^2 \varphi, \beta^2) \\ &= \beta y R_{-1/2}(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{3}{2}; \beta^2 - y^2, \alpha^2 - x^2 + \beta^2 - y^2, \beta^2) \\ &= R_{-1/2}(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{3}{2}; y^{-2} - \beta^{-2}, y^{-2} - \beta^{-2} + \alpha^2 \beta^{-4}, y^{-2}). \end{aligned} \quad (4)$$

Because the b parameters are unequal the function has no permutation symmetry. Reduction to symmetric functions is accomplished by (9.3-3), but the complexity of the result [cf. (9.2-13)] does not recommend the practical use of elliptic arcs.

The perimeter S of the ellipse is

$$S = 4 \int_0^{\pi/2} (\alpha^2 \sin^2 \theta + \beta^2 \cos^2 \theta)^{1/2} d\theta = 2\pi R_E(\alpha^2, \beta^2), \quad (5)$$

as we see from Exercise 5.9-19 and (9.2-3) or from (9.2-14). A corresponding formula for an ellipsoid is given in the next example.

EXAMPLE 9.4-2 (Surface area of an ellipsoid) A vector surface element of the unit sphere has components $(l, m, n) \sin \theta d\theta d\varphi$, where $l = \sin \theta \cos \varphi$, $m = \sin \theta \sin \varphi$, $n = \cos \theta$. The unit sphere is deformed into the ellipsoid with semiaxes α, β, γ by the scale transformation $(x, y, z) \mapsto (\alpha x, \beta y, \gamma z)$, which carries the surface element of the sphere into the surface element $(\beta\gamma l, \gamma\alpha m, \alpha\beta n) \sin \theta d\theta d\varphi$ of the ellipsoid. (For example a surface element with its normal parallel to the z axis is stretched by a factor α in the x direction and β in the y direction.) The area S of the ellipsoid is

$$\begin{aligned} S &= 8 \int_0^{\pi/2} \int_0^{\pi/2} (\beta^2 \gamma^2 l^2 + \gamma^2 \alpha^2 m^2 + \alpha^2 \beta^2 n^2)^{1/2} \sin \theta d\theta d\varphi \\ &= 2B(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}) R_{1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; \beta^2 \gamma^2, \gamma^2 \alpha^2, \alpha^2 \beta^2) \\ &= 4\pi R_G(\beta^2 \gamma^2, \gamma^2 \alpha^2, \alpha^2 \beta^2) = 4\pi \alpha \beta \gamma R_G(\alpha^{-2}, \beta^{-2}, \gamma^{-2}), \end{aligned} \quad (6)$$

where we have used Exercise 5.9-20. Thus the symmetric standard function R_G defined in (9.2-1) is related to the area of an ellipsoid instead of the length of an elliptic arc. The area must be symmetric in α, β, γ because it is unchanged by permuting the semiaxes.

An ellipsoid in $\mathbb{R}^n$, $n \geq 2$, with semiaxes $\alpha_1, \dots, \alpha_n$ has a ratio of surface area to volume expressed by [see Carlson (1966)]

$$S/nV = R_{1/2}(\tfrac{1}{2}, \dots, \tfrac{1}{2}; \alpha_1^{-2}, \dots, \alpha_n^{-2}). \quad (7)$$

The volume V is calculated in Exercise 4.3-6. If $n = 2$, (7) reduces to (5), V being in this case the area $\pi\alpha\beta$ of the ellipse.

EXAMPLE 9.4-3 (*Capacity and potential of a charged conducting ellipsoid*) By introducing ellipsoidal coordinates we shall express the electrostatic potential and capacity of a charged conducting ellipsoid in terms of the standard function R_F . The equipotential surfaces are a family of ellipsoids confocal with the surface of the conductor. In the case of an ellipsoid of revolution R_F reduces to the elementary function R_C . In the case of an elliptic disk the reciprocal capacity reduces to a complete elliptic integral R_K .

The ellipsoidal coordinates λ, μ, ν of a point in $\mathbb{R}^3$ with Cartesian coordinates x, y, z are defined to be the three roots of the cubic equation in θ ,

$$\frac{x^2}{\theta + \alpha^2} + \frac{y^2}{\theta + \beta^2} + \frac{z^2}{\theta + \gamma^2} = 1, \quad (8)$$

where α, β, γ are fixed positive numbers. The graph of the left side as a function of θ shows that the roots are real. The largest root, say λ , is such that all three terms on the left side are positive. Thus the surfaces of constant λ are confocal ellipsoids which approach spherical shape at large distances from the origin. As $r \rightarrow \infty$, $r^2/\lambda \rightarrow 1$.

If the surface $\lambda = 0$ (an ellipsoid with semiaxes α, β, γ) is a conductor bearing a total charge Q , it must be an equipotential surface. To solve Laplace's equation with the boundary condition that the potential V is constant on the surface $\lambda = 0$, it is reasonable to try a solution that depends only on λ . For such a function Laplace's equation takes the form [Whittaker (1927, p. 552), Bateman (1932, p. 422)]

$$\frac{d}{d\lambda} [(\lambda + \alpha^2)(\lambda + \beta^2)(\lambda + \gamma^2)]^{1/2} \frac{dV}{d\lambda} = 0. \quad (9)$$

Two integrations yield

$$V(\lambda) = A \int_{\lambda}^{\infty} [(t + \alpha^2)(t + \beta^2)(t + \gamma^2)]^{-1/2} dt + B,$$

where A and B are constants. Using (8.1-3) and the boundary condition that $\lambda^{1/2}V \sim rV \rightarrow Q$ as $r \rightarrow \infty$, we find

$$V(\lambda) = QR_F(\lambda + \alpha^2, \lambda + \beta^2, \lambda + \gamma^2). \quad (10)$$

The electric capacity $C = Q/V(0)$ is given by

$$1/C = R_F(\alpha^2, \beta^2, \gamma^2). \quad (11)$$

For an ellipsoid of revolution with $\gamma = \beta$, (8) reduces to a quadratic equation of which λ is the larger root, and (10) reduces by (8.2-13) to

$$V(\lambda) = QR_C(\lambda + \alpha^2, \lambda + \beta^2). \quad (12)$$

For an elliptic disk with $\gamma = 0$, (11) reduces by (9.2-2) to

$$1/C = (\pi/2)R_K(\alpha^2, \beta^2). \quad (13)$$

Since $4\pi Q$ is the surface integral of the magnitude of the potential gradient, the capacity can be defined in terms of a solution of Laplace's equation without reference to electricity. The notion of capacity is thereby extended to surfaces in spaces of higher dimension. An ellipsoid in $\mathbb{R}^n$, $n \geq 3$, has capacity C represented by [see Carlson (1966)]

$$(n-2)/C = R_{1-n/2}(\frac{1}{2}, \dots, \frac{1}{2}; \alpha_1^2, \dots, \alpha_n^2). \quad (14)$$

The case $n = 4$ will figure in Exercise 9.8-2.

EXAMPLE 9.4-4 (Anharmonic oscillator) A classical particle of mass m and energy E oscillates anharmonically with potential energy

$$V(x) = \frac{1}{2}kx^2 + \beta x^4, \quad (15)$$

where k and β are positive constants. Let A be the amplitude of oscillation, defined by $V(A) = E$ and $A > 0$, and let the initial conditions at $t = 0$ be $x = 0$ and $dx/dt = \omega A$, where ω is a positive number satisfying $E = \frac{1}{2}m\omega^2 A^2$. Define $\sigma = \beta A^4/E$, a dimensionless measure of nonlinearity. Then the kinetic energy is

$$\frac{1}{2}m(dx/dt)^2 = E - V(x) = E(1 - x^2/A^2)(1 + \sigma x^2/A^2). \quad (16)$$

Solve for dt/dx and integrate by (8.1-2); thus

$$\omega t = R_F\left(\frac{A^2}{x^2} - 1, \frac{A^2}{x^2}, \frac{A^2}{x^2} + \sigma\right). \quad (17)$$

In a quarter period x increases from 0 to A . By (9.2-3) the period T is given by

$$\omega T = 2\pi R_K(1, 1 + \sigma). \quad (18)$$

EXAMPLE 9.4-5 (Simple pendulum) The nonlinear differential equation of the simple pendulum is $d^2\theta/dt^2 + \omega^2 \sin \theta = 0$, where θ is the angle from the vertical and ω is the angular frequency of small oscillations. At time $t = 0$ let $\theta = 0$ and $d\theta/dt = 2\omega k$, where k is a positive constant. A first integration yields the energy equation

$$\frac{1}{2}(d\theta/dt)^2 + 2\omega^2 \sin^2 \frac{1}{2}\theta = 2\omega^2 k^2, \quad (19)$$

and the substitution $\sin \frac{1}{2}\theta = kx$ changes this to

$$(dx/dt)^2 = \omega^2(1 - x^2)(1 - k^2x^2). \quad (20)$$

In physical terms x^2 is the ratio of potential energy to total energy, and k^2 is the ratio of total energy to the change of potential energy between $\theta = 0$ and $\theta = \pi$. (This change is a natural unit of energy for the pendulum.) Solving for dt/dx and integrating by (8.1-2) yields

$$\omega t = R_F(x^{-2} - 1, x^{-2} - k^2, x^{-2}). \quad (21)$$

In the oscillatory case ($k < 1$), x increases from 0 to 1 during a quarter period as θ increases from 0 to a maximum value $\varphi = 2 \arcsin k$. The period T is expressed by

$$\omega T = 4R_F(0, 1 - k^2, 1) = 2\pi R_K[\cos^2(\varphi/2), 1], \quad (22)$$

in agreement with Exercise 6.10-4. For small oscillations ($\varphi \sim 2k \rightarrow 0$) it follows that $\omega T \rightarrow 2\pi$, $x \sim \theta/\varphi$, and

$$\omega t \sim R_C(x^{-2} - 1, x^{-2}) = \arcsin x. \quad (23)$$

For large oscillations $\pi - \varphi \sim 2(1 - k^2)^{1/2} \rightarrow 0$ and

$$\omega T \sim 4 \log \frac{4}{\cos(\varphi/2)} \sim 4 \log \frac{8}{\pi - \varphi} \quad (24)$$

by Exercise 9.2-1. In the limiting case $k = 1$,

$$\omega t = R_C(x^{-2}, x^{-2} - 1) = \operatorname{arctanh} x = \operatorname{arctanh}(\sin \frac{1}{2}\theta). \quad (25)$$

If $k > 1$, then θ increases from 0 to π during a quarter period as x increases from 0 to $1/k$. The pendulum rotates end over end with period given by

$$\omega T = 4R_F(k^2 - 1, 0, k^2) = 2\pi R_K(k^2 - 1, k^2). \quad (26)$$

If (21) is regarded as a Schwarz–Christoffel mapping (8.2-7), the transition from oscillation to rotation occurs when two critical points of the mapping coincide. If $k < 1$, then $x = \operatorname{sn} \omega t$ by comparison of (21) and (8.2-15).

9.5 Landen's transformation

In 1775 Landen established a relation between a hyperbolic arc and two elliptic arcs which Legendre later recast as a quadratic transformation of his standard integrals. Successive transformations move an elliptic integral successively closer to an inverse circular or inverse hyperbolic function. Because the rate of convergence is quadratic, Legendre was able to devise an iterative algorithm in which the number of significant figures doubles in each cycle of iteration. With this method, implemented by herculean labors of hand calculation, he constructed a table of $F(\varphi, k)$ and $E(\varphi, k)$ containing over sixteen thousand entries with at least nine decimal places. The method of successive Landen transformations is still the one most frequently used for computing incomplete elliptic integrals.

The invariance of R_F under Landen's transformation implies, as we shall see, the invariance of R_K and R_C under the quadratic transformations which respectively generate the algorithm of the arithmetic-geometric mean (Example 6.10-2) and Borchardt's algorithm (Example 6.10-5). Since these transformations are special cases of (6.10-1) and (6.10-20), it is not surprising that Landen's transformation is a special case of a transformation with two free parameters [see Carlson (1967a)]:

$$R_t(\beta, \beta, \frac{1}{2} - \beta - t; x^2, y^2, z^2) = u^{1-2\beta} R_{\beta+t-1/2}(\beta + t + \frac{1}{2}, -t, -t; u^2, v^2, w^2), \quad (1)$$

$$u = \frac{1}{2}(x + y), \quad v + w = (z + x)^{1/2}(z + y)^{1/2}, \quad v - w = (z - x)^{1/2}(z - y)^{1/2}. \quad (2)$$

The relations inverse to (2) are

$$z = vw/u, \quad x + y = 2u, \quad x - y = (2/u)(v^2 - u^2)^{1/2}(w^2 - u^2)^{1/2}. \quad (3)$$

All quadratic transformations of ${}_2F_1$ can be deduced from (1) because the two independent transformations (6.10-1) and (6.10-20) follow from the special cases $z = 0$ and $z = y$ respectively (Exercise 9.5-1). The Landen transformation of R_F is the case $-t = \beta = \frac{1}{2}$,

$$R_F(x^2, y^2, z^2) = R_F(u^2, v^2, w^2). \quad (4)$$

If $z = 0$, (4) becomes the invariance relation (6.10-5) for R_K , and if $z = y$, it is the invariance relation (6.10-23) for R_C . The Landen transformation of R_G can be deduced from (1) by setting $t = -\frac{3}{2}$ and $\beta = \frac{1}{2}$ (Exercise 9.5-5). We shall prove (1) only when $\beta = \frac{1}{2}$, since this case [Carlson (1964)] suffices for the Landen transformations of R_F and R_G . [For the transformation of R_H see Zill (1970).]

THEOREM 9.5-1 Let $1 - t \in U$ and $x, y, z \in \mathbb{R}_+$. Assume $xy + yz + zx > 0$, $(z + x)^{1/2} > 0$, $(z + y)^{1/2} > 0$, and define u, v, w by (2). Then

$$R_t(\frac{1}{2}, \frac{1}{2}, -t; x^2, y^2, z^2) = R_t(t + 1, -t, -t; u^2, v^2, w^2). \quad (5)$$

Remark By the permanence of functional relations (5) remains valid if $x, y, z \in \mathbb{C}_>$ and $u, v, w \in \mathbb{C}_>$, but it is not clear whether each of these conditions implies the other. The signs of $(z - x)^{1/2}$ and $(z - y)^{1/2}$ are immaterial because the right side of (5) is symmetric in v and w . A similar remark applies to the inverse transformation.

Proof Since both sides of (5) are holomorphic in t if $1 - t \in U$ by Theorem 6.8-2, the permanence of functional relations allows us to assume $\operatorname{Re} t < 0$. By (6.8-6),

$$R_t(\frac{1}{2}, \frac{1}{2}, -t; x^2, y^2, z^2) = -t \int_0^\infty (r + x^2)^{-1/2} (r + y^2)^{-1/2} (r + z^2)^t dr. \quad (6)$$

A new integration variable s is related to r by

$$r = s(s + xy)/(s + u^2), \quad s = \frac{1}{4}[(r + x^2)^{1/2} + (r + y^2)^{1/2}]^2 - u^2. \quad (7)$$

Adding z^2 to the first equation and differentiating the second yields

$$r + z^2 = (s + v^2)(s + w^2)/s + u^2, \quad dr/(r + x^2)^{1/2}(r + y^2)^{1/2} = ds/(s + u^2). \quad (8)$$

The right side of (6) becomes

$$-t \int_0^\infty (s + u^2)^{-t-1} (s + v^2)^t (s + w^2)^t ds = R_t(t + 1, -t, -t; u^2, v^2, w^2),$$

in which we have again used (6.8-6). $\square$

The left side of (4) is symmetric in x, y, z while the right side has formal symmetry only in x and y . Thus (4) contains three distinct transformations. For example if (x, y, z) is a permutation of $(1, 2, 3)$, there are three sets of values (u, v, w) corresponding to the three choices $z = 1$, $z = 2$, and $z = 3$. If $z = 2$, v and w are complex. This suggests that we discuss first the cases $z > x, y$ and $z < x, y$.

It follows from (2) that

$$\begin{aligned} 4(v^2 - u^2) &= [(z^2 - x^2)^{1/2} + (z^2 - y^2)^{1/2}]^2, \\ 4(w^2 - u^2) &= [(z^2 - x^2)^{1/2} - (z^2 - y^2)^{1/2}]^2. \end{aligned} \quad (9)$$

If $z > x, y$, we set

$$\begin{aligned}(x^2, y^2, z^2) &= (s_n^2, s_n^2 + c_n^2, s_n^2 + a_n^2), \\ (u^2, v^2, w^2) &= (s_{n+1}^2, s_{n+1}^2 + a_{n+1}^2, s_{n+1}^2 + c_{n+1}^2),\end{aligned}\quad (10)$$

where $a_m \geq c_m \geq 0$ and $s_m \geq 0$ for $m = n, n+1$. By (2), (4), and (9), $R_F(s_n^2, s_n^2 + c_n^2, s_n^2 + a_n^2)$ is independent of n if

$$\begin{aligned}2s_{n+1} &= s_n + (s_n^2 + c_n^2)^{1/2}, & 2a_{n+1} &= a_n + (a_n^2 - c_n^2)^{1/2}, \\ 2c_{n+1} &= a_n - (a_n^2 - c_n^2)^{1/2}.\end{aligned}\quad (11)$$

Multiplying the last two equations shows that $c_{n+1} = c_n^2/4a_{n+1}$. If we define $b_n = (a_n^2 - c_n^2)^{1/2}$, then a_{n+1} and b_{n+1} are arithmetic and geometric means of a_n and b_n . If (11) is iterated and if $b_0 \neq 0$, then a_n and b_n tend to a common limit M as $n \rightarrow \infty$ while c_n tends quadratically to 0 and s_n to a limit S . By continuity of R_F and (8.2-13),

$$R_F(s_n^2, s_n^2 + c_n^2, s_n^2 + a_n^2) \rightarrow R_F(S^2, S^2, S^2 + M^2) = R_C(S^2 + M^2, S^2).$$

The use of c_n instead of b_n avoids loss of significant figures in calculating s_{n+1} when $s_n \ll a_n$. We have now proved the following algorithm.

ALGORITHM 9.5-2 (*Ascending Landen transformations*) Let $s_0 \geq 0$ and $a_0 > c_0 > 0$. For every $n \in \mathbb{N}$ define

$$\begin{aligned}2s_{n+1} &= s_n + (s_n^2 + c_n^2)^{1/2}, & 2a_{n+1} &= a_n + (a_n^2 - c_n^2)^{1/2}, \\ c_{n+1} &= c_n^2/4a_{n+1}, \\ S &= \lim_{n \rightarrow \infty} s_n, & M &= \lim_{n \rightarrow \infty} a_n.\end{aligned}\quad (12)$$

Then

$$R_F(s_0^2, s_0^2 + c_0^2, s_0^2 + a_0^2) = R_C(S^2 + M^2, S^2) = (1/M) \operatorname{arcsinh}(M/S). \quad (13)$$

If $z < x, y$ we replace (10) by

$$\begin{aligned}(x^2, y^2, z^2) &= (t_n^2, t_n^2 - c_n^2, t_n^2 - a_n^2), \\ (u^2, v^2, w^2) &= (t_{n+1}^2, t_{n+1}^2 - a_{n+1}^2, t_{n+1}^2 - c_{n+1}^2).\end{aligned}\quad (14)$$

From (2), (4), and (9) it follows that $R_F(t_n^2, t_n^2 - c_n^2, t_n^2 - a_n^2)$ is independent of n if a_{n+1} and c_{n+1} are again given by (11) while

$$2t_{n+1} = t_n + (t_n^2 - c_n^2)^{1/2}. \quad (15)$$

ALGORITHM 9.5-3 (*Descending Gauss transformations*) Let $t_0 \geq a_0 > c_0 > 0$. For every $n \in \mathbb{N}$ define

$$\begin{aligned} 2t_{n+1} &= t_n + (t_n^2 - c_n^2)^{1/2}, & 2a_{n+1} &= a_n + (a_n^2 - c_n^2)^{1/2}, \\ c_{n+1} &= c_n^2/4a_{n+1}, \\ T &= \lim_{n \rightarrow \infty} t_n, & M &= \lim_{n \rightarrow \infty} a_n. \end{aligned} \quad (16)$$

Then

$$R_F(t_0^2 - a_0^2, t_0^2 - c_0^2, t_0^2) = R_C(T^2 - M^2, T^2) = (1/M)\arcsin(M/T). \quad (17)$$

To calculate $R_F(X, Y, Z)$ assume $X < Y < Z$ by symmetry. (If the inequalities were not strict, R_F would reduce immediately to R_C .) Compute initial values such as $a_0 = (Z - X)^{1/2}$, using Algorithm 9.5-2 if $2Y < X + Z$ and Algorithm 9.5-3 if $2Y > X + Z$ to make $c_0^2 \leq \frac{1}{2}a_0^2$. (Since the last inequality implies $c_4/a_4 < 5 \times 10^{-11}$ and $c_5/a_5 < 6 \times 10^{-22}$, the convergence is very fast.) Use the recurrence relations (12) or (16) until c_n is negligible, and then approximate M by a_n and S by s_n or T by t_n with negligible error. The slower algorithm may be used as a check.

The s_n form an ascending sequence and the t_n a descending sequence, but the names of the algorithms originated from the increasing or decreasing property of a sequence of moduli starting with $k_0^2 = (Z - Y)/(Z - X)$.

Even the third transformation, in which z lies between x and y with the result that $w = \bar{v}$, is valuable when used backwards to change a pair of complex conjugate variables into positive variables. Alternatively, if $y = \bar{x}$, $x \in \mathbb{C}_+$, and $z > 0$, (2) and (4) imply

$$\begin{aligned} R_F(x^2, \bar{x}^2, z^2) &= R_F(u^2, v^2, w^2), \\ u &= \operatorname{Re} x, & v + w &= |z + x|, & v - w &= |z - x|. \end{aligned} \quad (18)$$

Algorithms for computing R_G are discussed by Carlson (1965a). For R_E see Exercise 9.5-4.

9.6 The duplication theorem

Fagnano discovered in 1718 that an arc of Bernoulli's lemniscate can be doubled with ruler and compass. This early theorem on elliptic integrals led Euler to the addition theorem (Section 9.7) thirty-five years later. Although the duplication theorem will be proved here by a change of integration variable, it is obtained more laboriously when an ascending

Landen transformation precedes or follows a descending Gauss transformation. Thus it is another quadratic transformation of R_F into itself; it is more symmetrical than Landen's transformation but gives linear rather than quadratic convergence when used repeatedly. The resulting algorithm, which requires only rational operations and square roots, reduces to Borchardt's algorithm (Example 6.10-5) when two variables are equal. The extrapolation method used to accelerate the convergence of Borchardt's algorithm is again very effective, making the new algorithm competitive with those of Section 9.5.

THEOREM 9.6-1 Let $x, y, z \in \mathbb{C}_0 \cup \{0\}$ and assume that at most one of x, y, z is 0. Define $\lambda = x^{1/2}y^{1/2} + x^{1/2}z^{1/2} + y^{1/2}z^{1/2}$. Then

$$R_F(x, y, z) = 2R_F(x + \lambda, y + \lambda, z + \lambda). \quad (1)$$

Remark Note that $x + \lambda = (x^{1/2} + y^{1/2})(x^{1/2} + z^{1/2}) \in \mathbb{C}_0$ and similarly $y + \lambda, z + \lambda \in \mathbb{C}_0$.

Proof By Exercise 9.2-3,

$$R_F(x, y, z) = \frac{1}{2} \int_0^\infty [(s+x)(s+y)(s+z)]^{-1/2} ds. \quad (2)$$

Assume for the moment that $x, y, z \in \mathbb{R}_>$ and introduce the new integration variable

$$t = [(s+x)^{1/2} + (s+y)^{1/2}][(s+x)^{1/2} + (s+z)^{1/2}] - x. \quad (3)$$

(Inversely s is a rational function of t .) Expansion of the product shows that the right side is symmetric in x, y, z , and thus

$$\begin{aligned} (t+x)(t+y)(t+z) &= (X+Y)^2(X+Z)^2(Y+Z)^2, \\ X &= (s+x)^{1/2}, \quad Y = (s+y)^{1/2}, \quad Z = (s+z)^{1/2}. \end{aligned} \quad (4)$$

Combining this with

$$\frac{dt}{ds} = \frac{(X+Y)(X+Z)(Y+Z)}{2XYZ}, \quad (5)$$

we find

$$[(s+x)(s+y)(s+z)]^{-1/2} ds = 2[(t+x)(t+y)(t+z)]^{-1/2} dt. \quad (6)$$

Since $s = 0$ implies $t = \lambda$, substitution in (2) and use of Exercise 9.2-3 yield (1). By the permanence of functional relations we may now drop the assumption that $x, y, z \in \mathbb{R}_>$ because both sides of (1) are holomorphic in (x, y, z) on $\mathbb{C}_0^3$. Moreover, if exactly one of x, y, z tends to 0, both sides of (1) are continuous by Section 8.3. $\square$

To arrange (1) for successive applications, we absorb the factor 2 by homogeneity and set

$$(x', y, z) = (x_n^2, y_n^2, z_n^2),$$

$$(\tfrac{1}{4}(x + \lambda), \tfrac{1}{4}(y + \lambda), \tfrac{1}{4}(z + \lambda)) = (x_{n+1}^2, y_{n+1}^2, z_{n+1}^2).$$

Then $R_F(x_n^2, y_n^2, z_n^2)$ is independent of n if

$$x_{n+1}^2 = \tfrac{1}{4}(x_n + y_n)(x_n + z_n),$$

with similar equations for y_{n+1} and z_{n+1} . Since x_{n+1} is the geometric mean of the arithmetic means $\tfrac{1}{2}(x_n + y_n)$ and $\tfrac{1}{2}(x_n + z_n)$, it is evident [cf. (10) below] that x_n, y_n, z_n have a common limit L as $n \rightarrow \infty$. By continuity of R_F ,

$$R_F(x_n^2, y_n^2, z_n^2) \rightarrow R_F(L^2, L^2, L^2) = 1/L.$$

This proves the following algorithm [Carlson (1970c)].

ALGORITHM 9.6-2 Let $x_0, y_0, z_0 \in \mathbb{C}_> \cup \{0\}$ and assume that at most one of them is 0. For every $n \in \mathbb{N}$ define three numbers in $\mathbb{C}_>$ by

$$\begin{aligned} x_{n+1} &= \tfrac{1}{2}(x_n + y_n)^{1/2}(x_n + z_n)^{1/2}, & y_{n+1} &= \tfrac{1}{2}(y_n + x_n)^{1/2}(y_n + z_n)^{1/2}, \\ z_{n+1} &= \tfrac{1}{2}(z_n + x_n)^{1/2}(z_n + y_n)^{1/2}. \end{aligned} \quad (7)$$

Then x_n, y_n, z_n tend to a common limit L as $n \rightarrow \infty$, and

$$R_F(x_0^2, y_0^2, z_0^2) = 1/L. \quad (8)$$

In each cycle of iteration it suffices to extract the square roots of x_{n+1}^2 and y_{n+1}^2 because

$$z_{n+1} = 2x_{n+1}y_{n+1}/(x_n + y_n). \quad (9)$$

Since (7) implies

$$x_n^2 - y_n^2 = 4^{-n}(x_0^2 - y_0^2), \quad y_n^2 - z_n^2 = 4^{-n}(y_0^2 - z_0^2), \quad (10)$$

the error committed in approximating L by x_n, y_n , or z_n ultimately decreases by a factor of four in each cycle. The factor of four is improved to 16 if L is approximated by

$$\sigma_n = \tfrac{1}{3}(x_n + y_n + z_n), \quad (11)$$

for (10) and Exercise 9.6-1 imply

$$\frac{\sigma_n}{L} = 1 + \frac{4^{-2n-1}}{90L^4} [(x_0^2 - y_0^2)^2 + (x_0^2 - z_0^2)^2 + (y_0^2 - z_0^2)^2] + O(4^{-3n}). \quad (12)$$

Further improvement from 16 to 64 or 256 is achieved by using (11) in (6.10-29) or (6.10-30).

If $y_0 = z_0$, then (10) implies $y_n = z_n$ for all $n \in \mathbb{N}$, and the algorithm reduces to Borchardt's algorithm (Example 6.10-5) by (8.2-13). Nearly complete integrals can be computed faster after applying (9.7-17).

9.7 The addition theorem

Two days before Christmas in 1751 Euler received the collected works of Fagnano, including his duplication theorem of 1718 for an arc of Bernoulli's lemniscate. In the next year Euler discovered the addition theorem for lemniscatic arcs and later the addition theorem for the general elliptic integral of the first kind. If p is a polynomial of degree not exceeding four, this theorem asserts that the differential equation

$$\frac{d\lambda}{[p(\lambda)]^{1/2}} + \frac{d\mu}{[p(\mu)]^{1/2}} = 0 \quad (1)$$

has an algebraic solution of the form

$$Q(\lambda, \mu) = 0, \quad (2)$$

where Q is a polynomial in λ and μ of degree 2 in each variable. If (1) is integrated in terms of elliptic integrals, their arguments are connected by the biquadratic relation (2).

The theorem is verified by writing

$$Q(\lambda, \mu) = a(\mu)\lambda^2 + 2b(\mu)\lambda + c(\mu) = \alpha(\lambda)\mu^2 + 2\beta(\lambda)\mu + \gamma(\lambda), \quad (3)$$

where $a, b, c, \alpha, \beta, \gamma$ are quadratic polynomials. It follows from (3) and (2) that

$$\begin{aligned} \frac{1}{2} \frac{\partial Q}{\partial \lambda} &= a\lambda + b, & \frac{1}{2} \frac{\partial Q}{\partial \mu} &= \alpha\mu + \beta, \\ (a\lambda + b)^2 - b^2 + ac &= aQ = 0, & (\alpha\mu + \beta)^2 - \beta^2 + \alpha\gamma &= \alpha Q = 0, \end{aligned} \quad (4)$$

and hence

$$\frac{1}{2} \frac{\partial Q}{\partial \lambda} = (b^2 - ac)^{1/2} = [p(\mu)]^{1/2}, \quad \frac{1}{2} \frac{\partial Q}{\partial \mu} = (\beta^2 - \alpha\gamma)^{1/2} = [q(\lambda)]^{1/2}, \quad (5)$$

where $p = b^2 - ac$ and $q = \beta^2 - \alpha\gamma$ are quartic polynomials. Finally,

$$\begin{aligned} \frac{\partial Q}{\partial \lambda} d\lambda + \frac{\partial Q}{\partial \mu} d\mu &= 0, \\ [p(\mu)]^{1/2} d\lambda + [q(\lambda)]^{1/2} d\mu &= 0. \end{aligned} \quad (6)$$

If $Q(\lambda, \mu)$ is symmetric in λ and μ , then $a = \alpha$, $b = \beta$, $c = \gamma$, $p = q$, and (6) reduces to (1).

Determining the coefficients of Q from those of p is tedious in general [see Cayley (1895, Chapter XIV)]. We shall consider the special case

$$a(\mu) = \mu^2, \quad b(\mu) = -f\mu - 2g, \quad c(\mu) = f^2 - 4g(\mu + e), \quad (7)$$

where e, f, g are constants. The two terms containing g satisfy the requirement that $a\lambda^2 + 2b\lambda + c$ be symmetric in λ and μ , whence $a = \alpha$, $b = \beta$, $c = \gamma$. Then

$$p(\mu) = q(\mu) = b^2 - ac = 4g(\mu^3 + e\mu^2 + f\mu + g) \quad (8)$$

and

$$Q(\lambda, \mu) = a\lambda^2 + 2b\lambda + c = (\lambda\mu - f)^2 - 4g(\lambda + \mu + e). \quad (9)$$

By (1) and (2) the differential equation

$$(\lambda^3 + e\lambda^2 + f\lambda + g)^{-1/2} d\lambda + (\mu^3 + e\mu^2 + f\mu + g)^{-1/2} d\mu = 0 \quad (10)$$

has the algebraic solution

$$(\lambda\mu - f)^2 = 4g(\lambda + \mu + e). \quad (11)$$

This is a particular solution with integration constant chosen so that $\mu \rightarrow 0$ as $\lambda \rightarrow \infty$. To obtain μ as an explicit function of λ , one can either solve (11) or recall from (4) and (5) that $\alpha(\lambda)\mu + \beta(\lambda) = [q(\lambda)]^{1/2}$, whence

$$\mu = \lambda^{-2}[f\lambda + 2g + 2g^{1/2}(\lambda^3 + e\lambda^2 + f\lambda + g)^{1/2}]. \quad (12)$$

THEOREM 9.7-1 (Addition theorem) Let $x, y, z \in \mathbb{R}_+$ and assume that at most one of x, y, z is 0. Let $\lambda, \mu \in \mathbb{R}_>$ and assume

$$\lambda\mu - xy - xz - yz = 2(xyz)^{1/2}(\lambda + \mu + x + y + z)^{1/2}, \quad (13)$$

where the square roots are nonnegative. Then

$$R_F(x + \lambda, y + \lambda, z + \lambda) + R_F(x + \mu, y + \mu, z + \mu) = R_F(x, y, z). \quad (14)$$

Remark This is an addition theorem in the usual sense for the inverse function $\lambda = f(u)$, where $u = R_F(x + \lambda, y + \lambda, z + \lambda)$. If x, y, z, λ, μ are replaced by $x + v, y + v, z + v, \lambda - v, \mu - v$, (13) becomes an algebraic relation between $\lambda = f(u)$, $\mu = f(v)$, and $v = f(u + v)$. See Exercise 9.7-2.

Proof Let $e = x + y + z, f = xy + xz + yz, g = xyz$. Then (10) becomes
$$[(\lambda + x)(\lambda + y)(\lambda + z)]^{-1/2} d\lambda + [(\mu + x)(\mu + y)(\mu + z)]^{-1/2} d\mu = 0, \quad (15)$$
 and (13) is the branch of (11) on which λ and μ are positive. Since

$\lambda \rightarrow 0$ as $\mu \rightarrow \infty$, integration of (15) yields

$$\int_0^\lambda [(t+x)(t+y)(t+z)]^{-1/2} dt - \int_\mu^\infty [(t+x)(t+y)(t+z)]^{-1/2} dt = 0. \quad (16)$$

Setting

$$\int_0^\lambda = \int_0^\infty - \int_\lambda^\infty$$

and using Exercise 9.2-3, we obtain (14). $\square$

Related theorems for R_G and R_H are discussed by Zill (1970). If $\lambda = \mu = x^{1/2}y^{1/2} + x^{1/2}z^{1/2} + y^{1/2}z^{1/2}$, (13) is satisfied and (14) reduces to the duplication theorem (9.6-1). If $z = 0$, (14) and (13) reduce by (9.2-2) to

$$R_F(x + \lambda, y + \lambda, \lambda) + R_F(x + \mu, y + \mu, \mu) = (\pi/2)R_K(x, y), \quad \lambda\mu = xy. \quad (17)$$

If λ is close to 0, the first term on the left is a nearly complete integral while the second term has nearly equal arguments (in the sense that their ratios are close to 1) because μ is large. Therefore Algorithm 9.6-2 requires fewer cycles of iteration for the second term than the first. Since R_K can be computed by the algorithm of the arithmetic-geometric mean (Example 6.10-2), (17) saves time in computing nearly complete integrals.

9.8 A reduction theorem

The general elliptic integral of the first kind is

$$I = \int_y^x \frac{dt}{[(a + \alpha t)(b + \beta t)(c + \gamma t)(d + \delta t)]^{1/2}}. \quad (1)$$

The branch points of the integrand are the zeros $-a/\alpha, \dots, -d/\delta$ of the quartic polynomial. If $d \neq 0$ and $\delta \rightarrow 0$, the last zero tends to ∞ and the quartic becomes a cubic polynomial, which is conveniently regarded as a quartic with one zero at infinity. If the integral is to be well-defined, no branch point can be an interior point of the path of integration, assumed here to be the line segment $[x, y]$. Evaluation by the formulas of Section 8.1 expresses the integral in terms of R_K if both limits of integration are branch points, R_F if one limit is a branch point, and $R_{-1}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ if neither limit is a branch point. Aside from a few technical exceptions, only the first two cases are listed in present integral tables. In the last case reduction to standard functions is accomplished by splitting the integral in two parts, as in (8.5-4), each of which has a branch point as one limit of

integration. An explicit formula for $R_{-1}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ in terms of two functions R_F is given in Exercise 8.5-2.

A simpler and more symmetrical reduction to a single function R_F is achieved by a recent theorem [Carlson (1977a, b)] which is proved here by change of integration variable. For a proof using the addition theorem see Exercise 9.8-1. We begin with a convenient representation of $R_{-1}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and subsequently return to (1).

THEOREM 9.8-1 Let $A, B, C, D \in \mathbb{C}_> \cup \{0\}$ and assume that at most one of A, B, C, D is 0. Define

$$X = AB + CD, \quad Y = AC + BD, \quad Z = AD + BC, \quad (2)$$

and assume $X, Y, Z \in \mathbb{C}_> \cup \{0\}$. Then

$$\begin{aligned} \int_0^\infty [(s + A^2)(s + B^2)(s + C^2)(s + D^2)]^{-1/2} ds \\ = \int_0^\infty [(t + X^2)(t + Y^2)(t + Z^2)]^{-1/2} dt, \end{aligned} \quad (3)$$

$$R_{-1}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; A^2, B^2, C^2, D^2) = 2R_F(X^2, Y^2, Z^2). \quad (4)$$

Remarks By (6.8-6), (3) and (4) are equivalent. It is impossible for two of X, Y, Z to be 0, for relations like $X + Y = (A + D)(B + C)$ would then imply that two of A, B, C, D have sum 0. If exactly one of A, B, C, D is 0, (4) may be verified explicitly by applying (8.3-2) and (6.8-15) to the left side. If $A = B$ and $C = D$, (4) reduces to (6.9-12).

Proof Since both sides of (4) are holomorphic if $A, \dots, Z \in \mathbb{C}_>$ and continuous if at most one variable tends to 0, it suffices to prove equality when $A, B, C, D \in \mathbb{R}_>$. Let

$$\begin{aligned} t &= [(s + A^2)^{1/2}(s + B^2)^{1/2} + (s + C^2)^{1/2}(s + D^2)^{1/2}]^2 - (AB + CD)^2 \\ &= (\alpha\beta + \gamma\delta)^2 - X^2, \end{aligned} \quad (5)$$

where $\alpha, \beta, \gamma, \delta$ are defined by

$$\begin{aligned} \alpha(s) &= (s + A^2)^{1/2}, & \beta(s) &= (s + B^2)^{1/2}, \\ \gamma(s) &= (s + C^2)^{1/2}, & \delta(s) &= (s + D^2)^{1/2}. \end{aligned} \quad (6)$$

Expansion of the squares in (5) shows that the middle member is symmetric in A, B, C, D , and hence

$$(t + X^2)(t + Y^2)(t + Z^2) = (\alpha\beta + \gamma\delta)^2(\alpha\gamma + \beta\delta)^2(\alpha\delta + \beta\gamma)^2. \quad (7)$$

Differentiating (5), we have

$$\begin{aligned}\frac{dt}{ds} &= (\alpha\beta + \gamma\delta)(\alpha^{-1}\beta + \alpha\beta^{-1} + \gamma^{-1}\delta + \gamma\delta^{-1}) \\ &= \frac{1}{\alpha\beta\gamma\delta} (\alpha\beta + \gamma\delta)(\alpha\gamma + \beta\delta)(\alpha\delta + \beta\gamma).\end{aligned}\quad (8)$$

From (7) and (8) it follows that

$$\begin{aligned}&[(s + A^2)(s + B^2)(s + C^2)(s + D^2)]^{-1/2} ds \\ &= [(t + X^2)(t + Y^2)(t + Z^2)]^{-1/2} dt,\end{aligned}\quad (9)$$

and the limits of integration in (3) agree with (5). $\square$

The group S_4 of permutations of A, B, C, D has an invariant subgroup V called the Klein four-group, containing the identity and three products of two-cycles: $(AB)(CD)$, $(AC)(BD)$, and $(AD)(BC)$. (Geometrically V is the symmetry group of a rectangle with vertices A, B, C, D .) The permutations belonging to V leave unchanged the combinations which are denoted in (2) by X, Y, Z , while the factor group S_4/V is isomorphic to the group S_3 of permutations of X, Y, Z . It is V which makes S_4 solvable and in Galois theory allows the general quartic equation to be solved by radicals. Since S_n is not solvable if $n > 4$, one does not expect an analog of (3) in more than four variables.[†]

Returning to (1), wherein $\alpha, \beta, \gamma, \delta$ are constants not related to (6), we find from (8.1-2) that

$$\begin{aligned}I &= (x - y)P^{-1/2}R_{-1}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; A^2, B^2, C^2, D^2), \\ P &= (a + \alpha y)(b + \beta y)(c + \gamma y)(d + \delta y), \\ A^2 &= \frac{a + \alpha x}{a + \alpha y}, \quad B^2 = \frac{b + \beta x}{b + \beta y}, \quad C^2 = \frac{c + \gamma x}{c + \gamma y}, \quad D^2 = \frac{d + \delta x}{d + \delta y}.\end{aligned}\quad (10)$$

Henceforth we assume $x, y \in \mathbb{R}$ and $x > y$. If each linear factor in the denominator of (1) is strictly positive for $x \geq t \geq y$, then $A, B, C, D \in \mathbb{R}_>$ and we may apply (4) with

$$\begin{aligned}X &= \left[\frac{(a + \alpha x)(b + \beta x)}{(a + \alpha y)(b + \beta y)} \right]^{1/2} + \left[\frac{(c + \gamma x)(d + \delta x)}{(c + \gamma y)(d + \delta y)} \right]^{1/2}, \\ Y^2 - X^2 &= (A^2 - D^2)(C^2 - B^2) = (x - y)^2 P^{-1} (a\delta - d\alpha)(c\beta - b\gamma), \quad (11) \\ Z^2 - X^2 &= (A^2 - C^2)(D^2 - B^2) = (x - y)^2 P^{-1} (a\gamma - c\alpha)(d\beta - b\delta).\end{aligned}$$

[†] This argument was suggested by I. R. Hentzel.

Setting $(X, Y, Z) = (x - y)P^{-1/2}(U, V, W)$ and using the homogeneity of R_F , we find

$$\begin{aligned} \int_y^x [(a + \alpha t)(b + \beta t)(c + \gamma t)(d + \delta t)]^{-1/2} dt &= 2R_F(U^2, V^2, W^2), \quad x > y, \\ (x - y)U &= [(a + \alpha x)(b + \beta x)(c + \gamma y)(d + \delta y)]^{1/2} \\ &\quad + [(a + \alpha y)(b + \beta y)(c + \gamma x)(d + \delta x)]^{1/2}, \\ V^2 - U^2 &= (a\delta - d\alpha)(c\beta - b\gamma), \quad W^2 - U^2 = (a\gamma - c\alpha)(d\beta - b\delta). \end{aligned} \quad (12)$$

This result remains valid by continuity if one limit of integration is a branch point, and in this case one of the two square roots in the expression for U vanishes. Equations for Y, Z , and $X^2 - Y^2$ similar to those in (11) show that

$$\begin{aligned} (x - y)V &= [(a + \alpha x)(b + \beta y)(c + \gamma x)(d + \delta y)]^{1/2} \\ &\quad + [(a + \alpha y)(b + \beta x)(c + \gamma y)(d + \delta x)]^{1/2}, \\ (x - y)W &= [(a + \alpha x)(b + \beta y)(c + \gamma y)(d + \delta x)]^{1/2} \\ &\quad + [(a + \alpha y)(b + \beta x)(c + \gamma x)(d + \delta y)]^{1/2}, \\ V^2 - W^2 &= (a\beta - b\alpha)(c\delta - d\gamma). \end{aligned} \quad (13)$$

Therefore, if both limits of integration are branch points, exactly one of U, V, W is 0 and R_F reduces to R_K .

If $x \rightarrow \infty$ or $y \rightarrow -\infty$, the quantities U, V, W have finite limits and (12) still holds. If at least one branch point is real, x and y cannot both be infinite.

If two branch points are conjugate complex and the other two are real or infinite, then two of A, B, C, D are in $\mathbb{R}_>$ while the other two are conjugate complex in $\mathbb{C}_>$. The Landen transformation (9.5-18) will then lead to positive variables. For explicit formulas see Carlson (1977a).

If there are two pairs of conjugate complex branch points, then A, B, C, D form two conjugate complex pairs in $\mathbb{C}_>$ and X, Y, Z, U, V, W are real. However, (4) fails if one of X, Y, Z is negative, and hence (12) is valid if and only if a certain condition is satisfied [see Carlson (1977a) for details]. Expressed geometrically the condition requires that the sum of the angles subtended by the real interval of integration at the two branch points in the upper half-plane must not exceed π . A sufficient but not necessary condition is that the open interval of integration does not contain the point at which the real axis is intersected by the diagonals of the quadrilateral having the branch points as vertices. An interesting property of this point is deduced in Exercise 9.8-4.

NOTES

See the Notes at the end of Chapter 1 for the history of elliptic integrals and elliptic functions, and those at the end of Chapter 8 for theory and applications. Cayley (1895) and Cazenave (1969) treat the properties of Legendre's integrals in detail. Byrd (1971), Milne-Thomson (1964), and Tölke (1950, Volumes 3, 4) give formularies and integral tables, and Nellis (1966) gives an extension of Table 9.3-1 with short numerical tables of R_F and R_G . The theory of R_H is discussed by Zill (1970). Liouville (1834) proved that Legendre's integrals cannot be expressed in terms of elementary functions. More direct but less illuminating proofs of Theorem 9.7-1 are given by Zill (1970) and Carlson (1977b); the latter puts Theorems 9.6-1 and 9.8-1 in slightly more general form. Theorem 9.6-1 is useful in obtaining algebraic upper and lower approximations to R_F [see Carlson (1970d)]. Carlson (1976a) gives a proof of the Bartky transformation for numerical computation of R_L .

Regarding lemniscatic arcs and Fagnano's duplication theorem, see Exercises 4.2-8 and 8.3-7, Watson (1933), Siegel (1969, Chapter 1, Section 1), Carlson (1971a), and Todd (1975).

EXERCISES

9.2-1 Let $x, y \in \mathbb{R}_+$. As $x \rightarrow 0$ deduce from (9.2-10) that

$$R_K(x^2, y^2) \sim (2/\pi y) \log(4y/x).$$

Show that this is equivalent to (8.3-16).

9.2-2 Let $\sin \varphi_1 = k \sin \varphi$ and $k_1 = 1/k$. Use permutation symmetry to deduce the reciprocal-modulus transformation,

$$F(\varphi_1, k_1) = kF(\varphi, k), \quad kE(\varphi_1, k_1) = E(\varphi, k) + (k^2 - 1)F(\varphi, k).$$

9.2-3 Let $x, y, z, \rho \in \mathbb{C}_0$ and $\lambda \in \mathbb{R}_+$. Show that

$$R_F(x + \lambda, y + \lambda, z + \lambda) = \frac{1}{2} \int_{\lambda}^{\infty} [(t+x)(t+y)(t+z)]^{-1/2} dt,$$

$$R_H(x + \lambda, y + \lambda, z + \lambda, \rho + \lambda) = \frac{3}{4} \int_{\lambda}^{\infty} [(t+x)(t+y)(t+z)]^{-1/2} \frac{t - \lambda}{t + \rho} dt.$$

9.3-1 Legendre's third integral is

$$\Pi(\varphi, k, \alpha^2) = \int_0^{\varphi} (1 - \alpha^2 \sin^2 \theta)^{-1} (1 - k^2 \sin^2 \theta)^{-1/2} d\theta.$$

Verify the third equation of (9.2-11) and show that

$$R_H(x, y, z, \rho)$$

$$= \frac{3(z-x)^{-1/2}}{2(\rho-z)} \left\{ \rho \Pi \left[\arccos \left(\frac{x}{z} \right)^{1/2}, \left(\frac{z-y}{z-x} \right)^{1/2}, \frac{z-\rho}{z-x} \right] - zF \left[\arccos \left(\frac{x}{z} \right)^{1/2}, \left(\frac{z-y}{z-x} \right)^{1/2} \right] \right\}.$$

Both sides are symmetric in x, y, z .

9.3-2 The mutual inductance of two coaxial of radii a and b with centers a distance h apart is

$$M = \frac{2\pi ab}{c^2} \int_0^{2\pi} (h^2 + a^2 + b^2 - 2ab \cos \theta)^{-1/2} \cos \theta \, d\theta,$$

where the squared speed of light c^2 in the denominator corresponds to Gaussian units. Let $r_{\pm}^2 = h^2 + (a \pm b)^2$ denote the square of the maximum (+) or minimum (−) distance between the circles. Show that

$$M = 2(\pi a^2/c)(\pi b^2/c)R_{-3/2}(\frac{3}{2}, \frac{3}{2}; r_+^2, r_-^2).$$

As $r_- \rightarrow 0$, show that

$$M \sim (4\pi a/c^2)[\log(8a/r_-) - 2].$$

This can be used [see Smythe (1968, Section 8.10)] to find the self-inductance of a circular loop of thin wire.

9.3-3 Let $x, y, z \in \mathbb{C}_0$. Show that

$$\frac{2}{3}(z-x)(z-y)R_H(x, y, z, z) = (xy - xz - yz)R_F(x, y, z) + 2zR_G(x, y, z) - (xyz)^{1/2}.$$

9.3-4 Let $x, y, \rho \in \mathbb{C}_0$ and assume $xy/\rho \in \mathbb{C}_0$. Show that

$$R_L(x, y, \rho) + R_L(x, y, xy/\rho) = 2R_K(x, y).$$

9.4-1 Show that the arc of the hyperbola $x^2/\alpha^2 - y^2/\beta^2 = 1$ connecting the points $(\alpha, 0)$ and (x, y) has length

$$s = R_{-1/2}(\frac{1}{2}, -\frac{1}{2}, \frac{3}{2}; y^{-2} + \beta^{-2}, y^{-2} + \beta^{-2} + \alpha^2\beta^{-4}, y^{-2}).$$

9.4-2 A homogeneous magnetic ellipsoid with susceptibility χ and surface $x^2/\alpha^2 + y^2/\beta^2 + z^2/\gamma^2 = 1$ is placed in a previously uniform magnetic field B parallel to the x axis. The external field and the induced magnetization together produce a uniform field inside the ellipsoid with strength $B/(1 + L_x\chi)$. The demagnetizing factor is

$$L_x = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{x^2}{\alpha^2} \sin \theta \, d\theta \, d\varphi,$$

where x is a Cartesian coordinate of a point on the surface with spherical polar coordinates θ, φ . If V denotes the volume of the ellipsoid, show that

$$L_x = (V/4\pi)R_{-3/2}(\frac{3}{2}, \frac{1}{2}, \frac{1}{2}; \alpha^2, \beta^2, \gamma^2).$$

9.4-3 Let f be real on $\mathbb{R}_+$, assume that $g(r) = 4\pi \int_r^\infty f(t)t \, dt$ exists for every $r \geq 0$, and assume g is square-integrable on $\mathbb{R}_+$. Let $\alpha, \beta, \gamma \in \mathbb{R}_+$. An ellipsoidal charge distribution in $\mathbb{R}^3$ with charge density $\rho(x, y, z) = f[(x^2/\alpha^2 + y^2/\beta^2 + z^2/\gamma^2)^{1/2}]$ has electric self-energy

$$U = \frac{1}{2} \int_{\mathbb{R}^6} \frac{\rho(x, y, z)\rho(x', y', z')}{[(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}} \, dx \, dy \, dz \, dx' \, dy' \, dz'.$$

Show that

$$U = \frac{1}{2} (\alpha\beta\gamma)^2 R_F(\alpha^2, \beta^2, \gamma^2) \int_0^\infty [g(r)]^2 \, dr.$$

9.5-1 Deduce (6.10-1) and (6.10-20) from (9.5-1).

9.5-2 Let $x, y \in \mathbb{C}_>$. Show that the Landen transformation of R_E is

$$R_E(x^2, y^2) = 2R_E\left[\left(\frac{x+y}{2}\right)^2, xy\right] - xyR_K(x^2, y^2).$$

9.5-3 Using the notation of Example 6.10-2, define

$$A(n) = 2^{n+1}[R_E(x_n^2, y_n^2) - x_n^2 R_K(x_n^2, y_n^2)].$$

Show that $A(n) \rightarrow 0$ as $n \rightarrow \infty$ and that

$$A(n) - A(n+1) = 2^n(y_n^2 - x_n^2)/M, \quad A(0) = \frac{1}{M} \sum_{n=0}^{\infty} 2^n(y_n^2 - x_n^2).$$

9.5-4 Using the notation of Example 6.10-2, define

$$B = x_1^2 - \sum_{n=2}^{\infty} 2^{n-1}(x_n^2 - y_n^2).$$

Show that

$$R_E(x_0^2, y_0^2) = B/M.$$

Compute $R_E(1, 2) = 1.216007\dots$

9.5-5 Using the notation and assumptions of Theorem 9.5-1, show that the Landen transformation of R_G is

$$2R_G(x^2, y^2, z^2) = 4R_G(u^2, v^2, w^2) - xyR_F(x^2, y^2, z^2) - z.$$

9.5-6 Use Algorithm 9.5-3 to compute $R_F(1, 1.5, 2) = 0.826017876\dots$

9.6-1 Let $x, y, z \in \mathbb{C}_>$ and $\varepsilon = |1 - x/z| + |1 - y/z|$. As $\varepsilon \rightarrow 0$ show that

$$\frac{1}{3}(x+y+z)R_F(x^2, y^2, z^2) = 1 + \frac{(x-y)^2 + (x-z)^2 + (y-z)^2}{10(x+y+z)^2} + O(\varepsilon^3).$$

9.6-2 Show that (9.6-7) implies

$$x_n = \frac{x_{n+1}y_{n+1}}{z_{n+1}} + \frac{x_{n+1}z_{n+1}}{y_{n+1}} - \frac{y_{n+1}z_{n+1}}{x_{n+1}}$$

and two similar equations obtained by permuting x, y, z .

9.7-1 Show that the solution (9.7-11) of the differential equation (9.7-10) can be written in the form

$$[(\lambda^3 + e\lambda^2 + f\lambda + g)^{1/2} - (\mu^3 + e\mu^2 + f\mu + g)^{1/2}]^2 = (\lambda - \mu)^2(\lambda + \mu + e).$$

9.7-2 Let $x, y, z, \lambda, \mu, v \in \mathbb{R}_+$ and assume that at most one of $x+v, y+v, z+v$ is 0. Assume further that $\lambda > v, \mu > v$, and

$$v = \left\{ \frac{[(\lambda+x)(\lambda+y)(\lambda+z)]^{1/2} - [(\mu+x)(\mu+y)(\mu+z)]^{1/2}}{\lambda - \mu} \right\}^2 - \lambda - \mu - x - y - z.$$

Show that

$$R_F(x + \lambda, y + \lambda, z + \lambda) + R_F(x + \mu, y + \mu, z + \mu) = R_F(x + v, y + v, z + v).$$

9.7-3 Show that (9.7-12) can be written in the form

$$\mu = \lambda^{-2}[(\lambda^3 + e\lambda^2 + f\lambda + g)^{1/2} + g^{1/2}]^2 - \lambda - e.$$

9.7-4 Show that (9.7-12) can be written in the form

$$\mu(\lambda^3 + e\lambda^2 + f\lambda + g)^{1/2} - \lambda(\mu^3 + e\mu^2 + f\mu + g)^{1/2} = (\lambda - \mu)g^{1/2}.$$

9.7-5 Show that the first equation in Exercise 9.7-2 can be written in the form

$$\begin{vmatrix} [(\lambda + x)(\lambda + y)(\lambda + z)]^{1/2} & \lambda & 1 \\ [(\mu + x)(\mu + y)(\mu + z)]^{1/2} & \mu & 1 \\ -[(v + x)(v + y)(v + z)]^{1/2} & v & 1 \end{vmatrix} = 0.$$

9.8-1 Prove (9.8-4) by using the addition theorem to combine the two terms on the right side of the equation in Exercise 8.5-2. Conversely, prove the addition theorem from (9.8-4) and Exercise 8.5-2.

9.8-2 Show that an ellipsoid in $\mathbb{R}^4$ with semiaxes A, B, C, D has the same capacity as an ellipsoid in $\mathbb{R}^3$ with semiaxes $AB + CD, AC + BD, AD + BC$.

9.8-3 Let $a, c, \alpha, \gamma \in \mathbb{R}$ and $0 \leq y < x \leq \infty$. Assume $at^2 + c$ and $\alpha t^2 + \gamma$ are strictly positive if $y < t < x$. Show that

$$\int_y^x [(at^2 + c)(\alpha t^2 + \gamma)]^{-1/2} dt = R_F(U^2, U^2 + a\gamma, U^2 + c\alpha),$$

$$(x^2 - y^2)U = x(ay^2 + c)^{1/2}(\alpha y^2 + \gamma)^{1/2} + y(ax^2 + c)^{1/2}(\alpha x^2 + \gamma)^{1/2}.$$

9.8-4 Let $p, q \in \mathbb{C} - \mathbb{R}$ and let r be the point at which the real axis is intersected by the diagonals of the quadrilateral with vertices $p, \bar{p}, q, \bar{q}$, where an overbar denotes complex conjugation. Show that r divides the real line so that the integrals of $|t - p|^{-1}|t - q|^{-1}$ over the two half-lines are equal. (Cf. Exercise 8.1-4.)

9.8-5 Let $x, y, z \in \mathbb{C}_>$. Define $X = xy + z^2$, $Y = (x + y)z$, and assume $X, Y \in \mathbb{C}_>$. Show that

$$R_{-1}(\tfrac{1}{2}, \tfrac{1}{2}, 1; x^2, y^2, z^2) = 2R_C(X^2, Y^2).$$

Show further that this result agrees with Exercise 8.5-1. The case $x = y$ is (6.9-12).

9.8-6 Let $a \in \mathbb{R}_>$. Assume that the quartic polynomial $Q(s) = as^4 + bs^3 + cs^2 + ds + e$ is strictly positive if $s > 0$ and either is positive or has a simple zero at $s = 0$. Define the cubic polynomial $C(t) = t^3 + ft^2 + gt + h$, where $f = c + 6a^{1/2}e^{1/2}$, $g = bd + 4ca^{1/2}e^{1/2} + 8ae$, and $h = (be^{1/2} + da^{1/2})^2$. (The two polynomials have the same invariants.) Show that

$$\int_0^\infty [Q(s)]^{-1/2} ds = \int_0^\infty [C(t)]^{-1/2} dt.$$